

(2) $u_r \in W \dots 0 \quad ? \quad \sim, \quad 11 \sim$

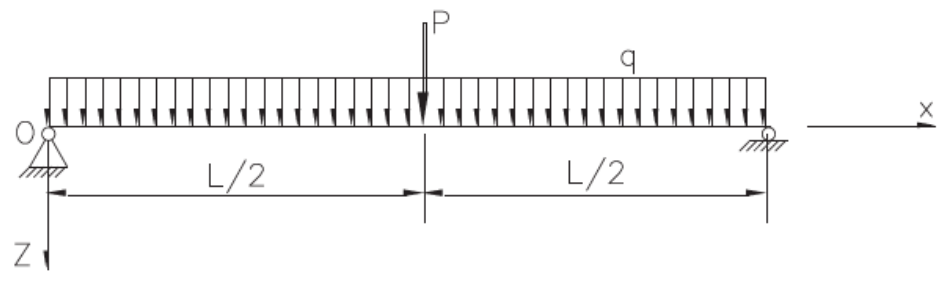
2.已知应变张量:

$$\varepsilon_{ij} = \begin{bmatrix} -3a & -a & 0 \\ -a & -2a & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

B "r(1) k ~ ;(2) k ~ ;(3)A ~ =G

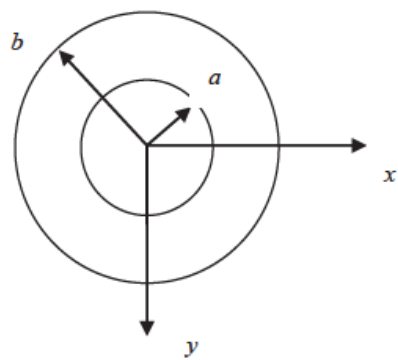
9" A 1`N"" ~ i50 6-
 11^(13 6) +.1 p / j j 0 1 y Z M 1 EF""Kfl LJ " j w39SEΔ L]
 ,P +X`B +X Y ~'(Rayleigh-Ritz) # F65 mEl1(Galerkin)# ~ ~ ... # F9 0/ ~ -
 "r P - W1~

$$/j^{\sim} P - w(\xi) \dot{=} a_1 \xi(1-\xi) + a_2 [\xi(1-\xi)]^2 \sim \text{ " } f \xi = x/L_1 \sim$$



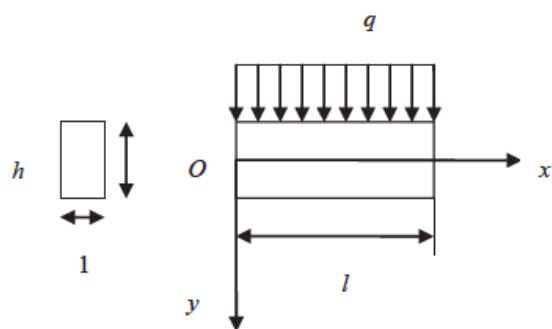
.1

21~10 6 ~ - A æ >~φ:=\frac{M}{2\pi}\theta %C^a B 3 ... Ø 0 ; ~ " f " r * () ~ , . ~ , 6G 1
 z \cdot ja^{\sim} " F z b , j 6) 1 + O ¶ :E(+~.2)^{\sim} " B " r * E Ø |^{\sim} :, \cdot Mf . / \mathfrak{J} \{



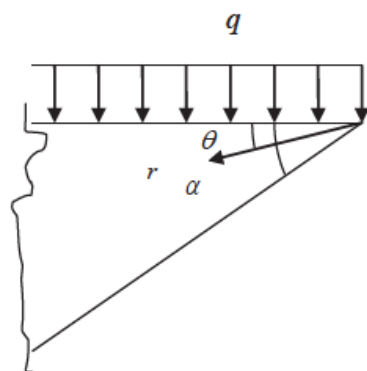
.2

3、(13 分)如图 3 所示，一长为 l ，高为 h 的矩形截面梁，受均布荷载 q 作用，试求该梁的挠度方程 $w(x)$ 。



.3

14 题：已知梁的挠度方程为 $w(x) = \frac{1}{6}qx^3 - \frac{1}{24}qx^4$ ，试求该梁的转角方程 $\theta(x)$ 和弯矩方程 $M(x)$ 。



.4

同济大学本科课程期终考试统一命题纸 A S 7 1 x

1\ 0 0

0° ° M N ~ ~ A j B N " ! . ~ i ' X B ' N " K I B ~ X h ~ ~ ' !] ? N " 6
 9 Z k ~ , Ø A 1 ° = (h ~ ~ + - \$, . 1 ~ ~ ~
 0 ? 7 - ~ * E y E > Ø 0 ; ... M 1 E Ø + | ~ &
 E /) 0 , . } 0 +) ~ , . 0 (° ... _ E 1 Ø ~ , i ~ -
 0 W ! ~ _ k ~ 1 ~ i ~ ~ ~
 f M ~ , L N " , . (' - _ (X Ø 0 Ø A , T j Z Ø A , ~ i j ~ (~
 0 W ~ ,) ~ f M : , . ! ~ , j M & 1 ~ ~ ~ h ~ -
 i W p E Ø + | : , . L] 9 S E > 9 ' S 5 \$ ~ * 6 n f i 4 * 6 E Ø + | &
 + X ~ , - > / j , . ~ , f 6 S Ø Ø ~ v = 0 ° % B Ø 3 Ø Ø ; ~ i ~ -
 4 E 1 F > , . M L * N \ , Ø ; æ = , j M & * \ Z . G , 1 ^ ~ ~ ... w j Ø +
 Z 1 ~ ~ h ~ -
 F 7 - , . } 0 + 1 N « % C a " . M , . E Ø + | & 1 ~ ~ ~ h ~ -
 ,) 0 ... L f P G , . (' - E - _ , 1 ~ ~ ~ 1 ~ -
) 1 ^ J F F J i W + ~ ? 5 * , . ~ 6 G ? - % C a B 3 Ø 0 ; a " r *
 0 + 6 G 1 ~ ~ h ~ -
 9 / Z " 9 J } 0 + ~ " I (+ ? & Ø 4 , . s G - " 9 J
 *) 0 W ~ , . f M ... @ T Z Ø ~ 1 , k i Ø ~ A
) + ? i W ~ " ~ , k Ø A ... h ~ - k Ø A 0 8 \$ 1 ~ ~ ~

... ° 6 N ~ ~ ; 20 6 ~ " ! ? N " 6 -

1.

(1) $Y+X \neq 0; \sigma_n = n \cdot \sigma \cdot n = \sigma_y n_i n_j \sim \epsilon$

$$\sigma_n = (a+b)\cos^2 \theta + (a-b)\sin^2 \theta = a+b\cos 2\theta$$

$$\% Y+X^2+\tau_n^2=(\sigma \cdot n) \cdot (\sigma \cdot n)=(a+b)^2 \cos ^2 \theta+(a-b)^2 \sin ^2 \theta \sim \epsilon$$

$$\tau_n^2=(a+b)^2 \cos ^2 \theta+(a-b)^2 \sin ^2 \theta-(a+b \cos 2 \theta)^2=b^2 \sin ^2 2 \theta \quad (6 \text{ 分})$$

(2) $+a \sim , P G \sigma , \sim , 6 G j M 3 \sim " Z \text{ !! } \sim , w j k \sim i \sim i+a 0 W k \sim$
 $j \sigma_x = a+b>0 \sim " 0 ? k \sigma_y = a-b<0 \sim " ! 0 W 0 ? \sim , 6 [j$

$$\tau_{\max }=b \ldots \tau_{\min }=-b \sim " \neq p) \sim , . Z 6 M / \# . \dagger G \$ > x E \geq \ldots y E / ; i$$

$$@ \sim \sim \sim =A \quad \emptyset \quad A 5 \sim \sim j \quad (4 \text{ 分})$$

2.

(1)由

$$(\boldsymbol{\varepsilon}-\boldsymbol{\varepsilon}I)=0$$

$$\partial_{\alpha}^{\beta}=\delta_{\alpha}^{\beta}-\delta_{\alpha}^{\beta}$$

$$\varepsilon_1=\frac{-5-\sqrt{5}}{2}a,\varepsilon_2=\frac{-5+\sqrt{5}}{2}a,\varepsilon_3=0\tag{4-6}$$

$$(2)\;k\sim\varnothing A$$

$$_{+a}\varepsilon_1,n_in_i=1\cdot n_1$$

$$_{+a}\varepsilon_2,n_in_i=1\cdot n_2$$

$$_{+a}\varepsilon_3,n_in_i=1\cdot n_3\tag{4-6}$$

$$(3)\; \widetilde{\varphi}=\varphi_{G,j}$$

$$J_1=-5a,J_2=5a^2,J_3=0\tag{2-6}$$

$$9^\circ\text{A}~1^\circ\text{N}^{\sim}\sim_{\mathfrak{i}}50\quad\mathfrak{E}^-$$

$$11\,\hat{G}^{\quad}+\mathfrak{X}^{\quad}Y^{\quad}\sim^{\quad}(\text{Rayleigh-Ritz})\,\#$$

$$\widetilde{\mathcal{T}}^{\sim-}\mathcal{A}^{\quad}\mathcal{P}^{\quad}-\mathfrak{W}(\xi)=a_1\xi(1-\xi)+a_2[\xi(1-\xi)]^2\sim^{\quad}\mathfrak{f}^{\quad}\mathfrak{E}^{\quad}=x/L\,\mathfrak{I}^{\sim}$$

$$\widetilde{\mathcal{Z}}^{\sim-}\quad\varnothing+\mathbb{E}\varnothing+\mathfrak{I}^{\quad}\&\sim$$

$$x=0\quad&\sim\mathfrak{E}=0\quad&\mathfrak{W}=0\sim^{\quad}\%~\mathbb{C}^{\mathfrak{A}}$$

$$x=L\quad&\sim\mathfrak{E}=1\quad&\mathfrak{W}=0\sim^{\quad}\%~\mathfrak{I}^{\sim}\tag{2-6}$$

$$\widetilde{\mathcal{Z}}^{\sim-}\quad\mathfrak{r}^{\quad}-,\cdot k\quad\mathfrak{H}$$

$$\frac{d\xi}{dx}=\frac{1}{L}$$

$$\frac{dw}{dx}=\frac{dw}{d\xi}\frac{d\xi}{dx}=\frac{1}{L}[a_1(1-2\xi)+2a_2\xi(1-\xi)(1-2\xi)]$$

$$\frac{d^2w}{dx^2}=\frac{d}{d\xi}(\frac{dw}{dx})(\frac{d\xi}{dx})=\frac{1}{L^2}(-2a_1+2a_2-12a_2\xi+12a_2\xi^2)$$

$$\begin{aligned}\Pi\sim&\frac{1}{2}\int_0^L EI(\frac{d^2w}{dx^2})^2\,dx-\int_0^L qwdx-Pw|_{x=\frac{L}{2}}\\&=\frac{EI}{L^3}(4a_1\sim\frac{4}{5}a_2)-qL(\frac{1}{6}a_1+\frac{1}{30}a_2)-P(\frac{1}{4}a_1+\frac{1}{16}a_2)\end{aligned}\tag{6-6}$$

$$\widetilde{\mathcal{Z}}^{\sim-}\quad9G+\mathfrak{X}^{\quad}Y^{\quad}\quad\sim^{\quad}(\text{Rayleigh-Ritz})\,\#\quad\sim$$

$$\frac{\partial \Pi}{\partial a_1}=0\text{ , 即 }\frac{4EI}{L^3}a_1=\frac{1}{6}qL+\frac{1}{4}P$$

④ 由

$$a_1=\frac{L^3}{48EI}(2qL+3P)$$

$$\frac{\partial \Pi}{\partial a_2}=0\text{ 得 }\frac{4EI}{5L^3}a_2=\frac{1}{30}qL+\frac{1}{16}P$$

⑤ 由

$$a_2=\frac{5L^3}{4EI}(\frac{1}{30}qL+\frac{1}{16}P)$$

将 P、F 代入式(1)得

$$w=\frac{qL^4}{24EI}[\xi(1-\xi)(1+\xi-\xi^2)]+\frac{PL^3}{64}[\xi(1-\xi)(4+5\xi-5\xi^2)]\qquad (5)$$

2.1

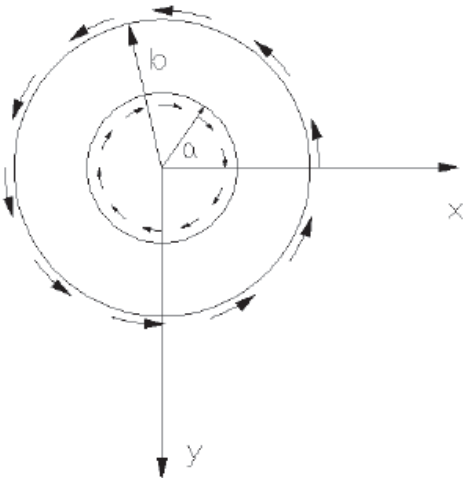
$$\varphi_1=\frac{M}{2\pi}\theta\qquad C\in B\cup\emptyset;$$

$$\varphi_2=\frac{M}{2\pi}\theta\qquad C\in B\cup\emptyset;$$

$$\sigma_r=\sigma_\theta=0,\tau_{r\theta}=\frac{M}{2\pi}\frac{1}{r^2}$$

$$E\varnothing r=a\qquad \tau_{r\theta}=\frac{M}{2\pi}\frac{1}{a^2}$$

$$E\varnothing r=b\qquad \tau_{r\theta}=\frac{M}{2\pi}\frac{1}{b^2}$$



3、

(1) 将 $\varphi = Ay^5 + Bx^2y^3 + Cy^3 + Dx^2 + Ex^2y$ B3 ... Ø þ

$$\nabla^2 \nabla^2 \varphi = 0$$

$$24By + 120Ay = 0$$

$$\therefore B = -5A \quad \& \quad \ddot{\varphi} \quad \text{j} \quad \sim \quad , \quad - \quad \sim \quad "$$

$$\therefore \varphi = Ay^5 - 5Ax^2y^3 + Cy^3 + Dx^2 + Ex^2y \quad (3 \quad 6)$$

$$\sim 2 \quad \sigma_x = \frac{\partial^2 \varphi}{\partial y^2} = 20Ay^3 - 30Ax^2y + 6Cy$$

$$\sigma_y = \frac{\partial^2 \varphi}{\partial x^2} = -10Ay^3 + 2D + 2Ey$$

$$\tau_{xy} = -\frac{\partial^2 \varphi}{\partial x \partial y} = 30Axy^2 - 2Ex \quad (3 \quad 6)$$

$$\sim 3 \quad \sim \quad E \quad \emptyset \quad + \quad \&$$

$$y = -h/2: \sigma_y = -q, \tau_{xy} = 0$$

$$y = h/2: \sigma_y = 0, \tau_{xy} = 0$$

$$x = 0: \int_{-\frac{h}{2}}^{\frac{h}{2}} \tau_{xy} dy = 0$$

$$\int_{-\frac{h}{2}}^{\frac{h}{2}} \sigma_x dy = 0$$

$$\int_{-\frac{h}{2}}^{\frac{h}{2}} \sigma_x y dy = 0$$

$$6\sigma_x \perp \hat{\sigma}_y \quad : \quad F \quad E \quad \emptyset \quad + \quad | \quad \& \quad @ \quad * \quad 43+ \quad (5 \quad 6)$$

$$\sim 4 \quad \sim \quad 6 \quad .$$

$$\sigma_x = \frac{4q}{h^3}y^3 - \frac{6q}{h}x^2y - \frac{3q}{5h}y$$

$$\sigma_y = -\frac{q}{2} \left(1 - \frac{3y}{h} + \frac{4y^3}{h^3} \right) \quad (2 \quad 6)$$

$$\tau_{xy} = \frac{6q}{h^3}xy^2 - \frac{3}{2} \frac{q}{h}x$$

4、

检查该应力函数满足双调和方程

(3 分)

在已知应力函数的情况下应力表达式为

$$\sigma_r = A - B \cos 2\theta - C \sin 2\theta + D\theta$$

$$\sigma_\theta = A + B \cos 2\theta + C \sin 2\theta + D\theta$$

(4 分)

$$\tau_{r\theta} = -\frac{1}{2}(-2B \sin 2\theta + 2C \cos 2\theta + D)$$

在 $\theta = 0$ 及 $\theta = \alpha$ 处

$$\sigma_\theta = 0, \sigma_\theta = -q$$

$$\tau_{r\theta} = 0, \tau_{r\theta} = 0$$

$$\sigma_\theta = \alpha, \sigma_\theta = 0$$

$$\tau_{r\theta} = \alpha, \tau_{r\theta} = 0$$

在 $\theta = 0$ 及 $\theta = \alpha$ 处

$$A + B = -q$$

$$-\frac{1}{2}(2C + D) = 0$$

(5 分)

$$A + B \cos 2\alpha + C \sin 2\alpha + D\alpha = 0$$

$$\frac{1}{2}(2B \sin 2\alpha - 2C \cos 2\alpha - D) = 0$$

在 $\theta = 0$ 及 $\theta = \alpha$ 处

$$A = \frac{q}{K} \left(\frac{1}{2} \tan \alpha - K \right), B = \frac{-q}{K} \frac{1}{2} \tan \alpha, C = \frac{1}{2} \frac{q}{K}, D = -\frac{q}{K}$$

(2 分)

$$K = \tan \alpha - \alpha$$